Assignment-6

RAVI

1Q. "People who often attend cultural activities, such as movies, sports events, and concerts, are more likely than their less cultured cousins to survive the next eight to nine years, even when education and income are taken into account, according to a survey by the University of Umea in Sweden "(American Health, April 1997, p. 20).

A)Can this claim be tested by conducting a randomized experiment? Explain

->No, this example cannot be tested as a randomized experiment because several reasons and events cannot be tested under a randomized experiment.

B) On the basis of the study that was conducted, can we conclude that attending cultural events causes people to be likely to live longer? Explain.

->No, we cannot conclude that attending cultural events causes people to be likely to live longer. Other factors may also have an effect.

C) The article notes that education and income were taken into account. How do you think they were taken into account?

-> people who have more income can attend most of the cultural events which are paid activities. Also, people who are educated have more interest to attend most cultural activities.

D) Give two other factors about the people surveyed that you think should also have been taken into account.

->Factors include any diseases that are present in the person, family history of activities, also bad habits such as sleeping too late, carelessness, etc may be taken into account.

2Q. Suppose a study, first asked people whether they meditate regularly and then measured their blood pressure. The idea would be to see if those who meditate have a lower blood pressure than those who do not do so.
a) Explain whether this would be an observational study or an experiment.

-> Given example is an observational study because the given blood pressure is concluded as per their existing meditation skills not randomly selected. Also, this effect on blood pressure cannot be experimented with in real life.
b) If it were found that meditators had lower than average blood pressure, can we conclude that meditation causes lower blood pressure? Explain

-> It depends on the people who meditate, there are people who have lower bp without meditation. So, it’s just an observational study we cannot state it as a causal effect.

3Q. Design an experiment to test whether an advertising campaign is effective in promoting the use of a given product.

-> we randomly select two towns and in town(A) we don't advertise and in Town(B) we advertise by taking this into account we compare both ideas and get the result.

4Q. Explain why a randomized experiment allows researchers to draw a causal conclusion whereas an observational study does not.

->A randomized experiment allows researchers to draw causal conclusions because in the randomized experiment we specifically select a sample of people from the overall population and give a causal inference to it. But in the observational study, we just take the given data which is not randomly selected and we analyze it.

4.6 (Data file: UN11) In the simple linear regression of log(fertility) on pctUrban using the UN11 data, the fitted model is log(fertility) = 1.501- 0.01pctUrban. Provide an interpretation of the estimated coefficient for pctUrban.

-> given equation is log(fertility) = 1.501- 0.01pctUrban

The coefficients:

intercept is positive and the value is 1.501 and the slope is negative and the value is -0.01.

the interpretation is when the pctUrban is zero then the value of log(fertility) rate is 1.501 and if we increase the variable pctUrban into 1 unit then we expect a 1% decrease in log(fertility).

4.7(Data file: UN11) Verify that in the regression log(fertility) $\sim$ log(ppgdp) + lifeExpF a 25% increase in ppgdp is associated with a 1.4% decrease in expected fertility.

->given regression function is log(fertility) $\sim$ log(ppgdp) + lifeExpF

In summary, we got the value of the coefficient for log(ppgdp) which is -0.06544.

This means a 1 unit increase in the ppgdp result in a decrease the fertility by -0.06544.

A 25% increase in the ppgdp. So, -0.06544*0.25=0.01635 which is 1.6%.

A 25% increase in the ppgdp resulted in a decrease the fertility by 1.6%.

4.9

A) given equation is : E(Salary/Sex) = 24697 - 3340Sex

The given β 0 is positive and the value is 24679 and the given β1 is negative and the value is -3340 which means when sex remains zero then the salary will be 24679 where 0 in our example is men and if the sex changes for 1 unit which means from male to female, then the salary will be decreased by 3340 units.

B) In the first equation there is only a relation between sex and salary. But here we have to consider then the relation of the year which is X2 in eq. so, the signs of sex change.